Final project script 2

Step 2. Measure features for each coin using a so-called "filter bank" of matching filters.

In step 1 we found the coins, but we still need to classify them. In step 3 we will perform classifcation. The classifier needs feature measurements f collect in this step. A "filter bank" of matching filters can be constructed, with distinct filters for quarters, nickels, and dimes. We can apply each of th coin in **msk_dil_erd**, the results of which generate 3 features that can be used by k-means. in the next step.

- In a separate MATLAB grader problem, you are asked to create function `MakeCircleMatchingFilter`. ***Complete that problem now and finished***. Embed the resulting function in the bottom of the script with the other Helper Functions
- Use the `MakeCircleMatchingFilter` function with the default variables provided in the script to create the dime, nickel, and quarter matc **nickelfilter**, and **quarterfilter**.
- For each *i*th coin centroid found in step 1, compute the *j*th (j=1, 2, and 3 should correspond to the dime, nickel, and quarter filters, respectiv computing the correlation between the matching filter and the local region of image pixels that fall within **filtsizeh** (half the matching filter w centroid. Store the result in **D(i,j).**

Script

Save Reset MATLAB Documentation (https://www.mathworks.com/help/)

```
% Define the filter size we will use in step 2:
filtsize = 85;

% Creating test image 'im' by splicing together two built in images.
% Also zero-padding (adding zeros around the border) with half the
% filter size (filtsize) we will use so that the filter could be
% centered on any actual image pixel, including those near the border.
% 'coins.png' contains bright nickels and dimes on a dark background
% 'eight.tif' contains dark quarters on a bright background, so we invert it
% to match 'coins.png'
im1 = imread('coins.png');
[r,c] = size(im1);
im2 = imread('eight.tif');
[r2,c2] = size(im2);
filtsizeh = floor(filtsize/2);
im = zeros(r+r2+filtsize,c+filtsize);
im(filtsizeh+1:filtsizeh+r+r2,filtsizeh+1:filtsizeh+c) = [im1;255-im2(:,1:c)];
[r,c] = size(im);
imagesc(im);colormap(gray);title('test image');axis equal;

% Initializing assessed/displayed variables as empty so that code is executable
msk=[]; msk_dil=[]; msk_dil_erd=[]; centroid=[]; component_size=[];

%%%%% 1. Localize the centroid of each coin
% Otsu threshold
msk = OtsuThreshold(im);
figure; imagesc(msk); colormap(gray); title('Otsu'); axis equal;

% Dilate 9x9
msk_dil = imdilate(msk,ones(9,9));
figure; imagesc(msk_dil); colormap(gray); title('Dilated'); axis equal;

% Erode 23x23
msk_dil_erd = imerode(msk_dil,ones(23,23));
figure; imagesc(msk_dil_erd); colormap(gray); title('Eroded'); axis equal;


% Connected components to get centroids of coins:
cc = bwconncomp(msk_dil_erd);
props_struct = regionprops(cc);
centroid = zeros(length(props_struct),2);
component_size = zeros(length(props_struct),1);
for i=1:length(props_struct)
    centroid(i,:) = round(props_struct(i).Centroid);
    component_size(i) = props_struct(i).Area;
end


%%%%% 2. Measure features for each coin using a bank of matching filters
% make matching filters to create features
% Define diameters to use for filters
dimediameter = 31;
quarterdiameter = 51;
nickeldiameter = 41;

% Initialize assessed variables
D=[]; nickelfilter = []; dimefilter = []; quarterfilter = [];

% Use the MakeCircleMatchingFilter function to create matching filters for dimes, nickels, and quarters
% (This is in a separate Matlab grader problem. Save your work,
%       complete the corresponding grader problem and embed the solution
%       in the helper function list below.)
[dimefilter, xc, yc] = MakeCircleMatchingFilter(dimediameter,filtsize);
[quarterfilter, xc, yc] = MakeCircleMatchingFilter(quarterdiameter,filtsize);
[nickelfilter, xc, yc] = MakeCircleMatchingFilter(nickeldiameter,filtsize);


figure;
subplot(1,3,1); imagesc(dimefilter); colormap(gray); title('dime filter'); axis tight equal;
subplot(1,3,2); imagesc(nickelfilter); colormap(gray); title('nickel filter'); axis tight equal;
subplot(1,3,3); imagesc(quarterfilter); colormap(gray); title('quarter filter'); axis tight equal;

% Evaluate each of the 3 matching filters on each coin to serve as 3 feature measurements
for i=1:size(centroid,1)
 for j=1:3
     filt = dimefilter;
     if j == 2
```

```
            filt = nickelfilter;
        elseif j == 3
            filt = quarterfilter;
        end
        x = centroid(i,1);
        y = centroid(i,2);
        D(i,j) = corr(filt(:), reshape(msk_dil_erd(y-filtsizeh:y+filtsizeh, x-filtsizeh:x+filtsizeh),[], 1));
    end
end
D

%%%%%%%%%%%%%%%%%%%% Helper Functions %%%%%%%%%%%%%%%%%%%%%


function [msk,thrsh] = OtsuThreshold(im)
```

Run Script

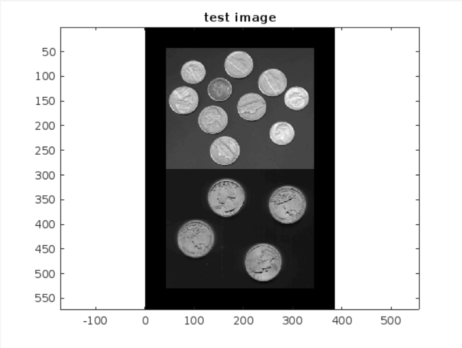
Assessment: All Tests Passed

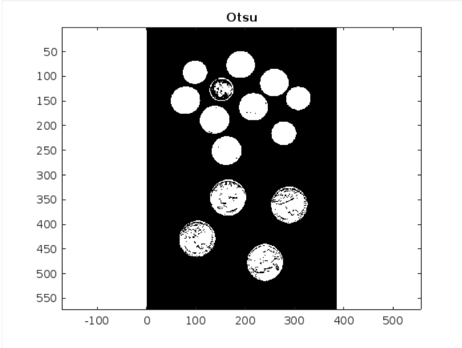
Submit

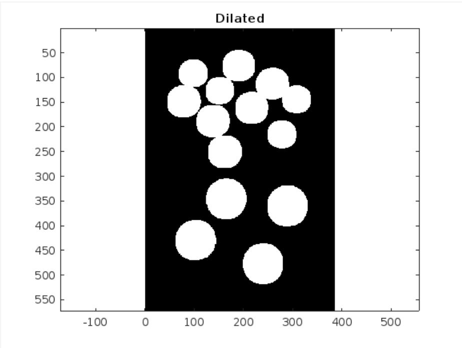
- Is nickelfilter correct?
- Is dimefilter correct?
- Is quarterfilter correct?
- Is D correct?

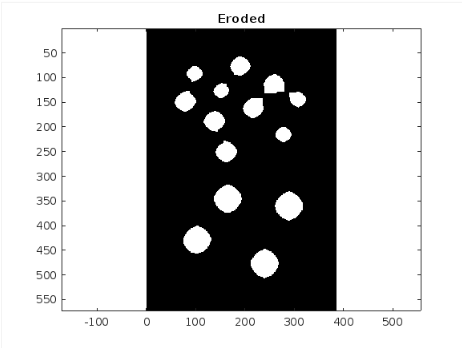
Output

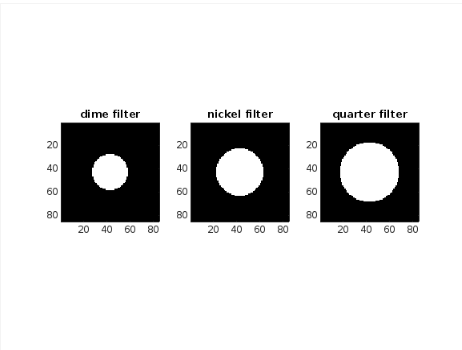
```
D =

    0.7237    0.9180    0.7013
    0.4948    0.6857    0.9160
    0.8499    0.6177    0.4439
    0.7157    0.9260    0.7044
    0.7801    0.5502    0.3817
    0.4932    0.6835    0.9138
    0.7196    0.9573    0.7453
    0.7392    0.8675    0.6413
    0.6415    0.8135    0.6234
    0.4858    0.6732    0.9001
    0.5697    0.7251    0.5555
    0.4857    0.6730    0.8998
    0.9610    0.7130    0.5333
    0.7024    0.5428    0.3613
```


test image
50
100
150
200
250
300
350
400
450
500
550
-100 0 100 200 300 400 500


Otsu
50
100
150
200
250
300
350
400
450
500
550
-100 0 100 200 300 400 500


Dilated
50
100
150
200
250
300
350
400
450
500
550
-100 0 100 200 300 400 500


Eroded
50
100
150
200
250
300
350
400
450
500
550
-100 0 100 200 300 400 500


dime filter
nickel filter
quarter filter
20
40
60
80
20 40 60 80